Profile-Driven Hardware Simulation for Organic Optoelectronic Edge Vision

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Abstract

We present a profile-driven behavioral simulation framework for organic optoelectronic compute-in-memory inference that maps literature device metrics to task-level vision accuracy. Across ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M on CIFAR-10/100 and Flowers-102, ADC resolution dominates under the canonical regime: below 6 bits causes abrupt collapse, while above 6 bits device-to-device variability becomes binding. Standard hardware-aware training overfits a single fixed mismatch map and collapses to 10.00% on fresh hardware instances, whereas Ensemble HAT—which resamples mismatch masks at each training epoch—recovers 86.37±1.54%. A literature-anchored OPECT case study reaches 88.53±0.08% zero-shot transfer without retraining. Under severe nonlinear write (NL=2.0), hardware-aware training with the revised gradient-scaling recipe recovers to the ∼80–82% band, leaving a residual gap of roughly 15 pp to the digital baseline that reflects a joint effect of non-linear write dynamics and fresh-instance transfer.

Keywords: organic optoelectronic compute-in-memory; hardware-aware training; vision transformers; edge vision; mixed-signal inference; profile-driven simulation.

1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications directly within memory arrays, eliminating the data-movement bottleneck that limits conventional digital systems[1–3]. Beyond conventional static random-access memory (SRAM) and emerging resistive memories, organic optoelectronic devices offer multilevel tunability, low-voltage operation, and inherent photosensitivity for direct optical signal processing[4–6]. Recent demonstrations span analog conductance states, long retention tails, and integrated optoelectronic synaptic arrays[5;7–11].

However, translating these device-level advances into system-level vision performance faces a critical methodological gap. The organic optoelectronic literature remains dominated by device-centric characterizations[12]. Network-level studies are typically limited to small benchmarks or simple perceptrons, while variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are either absent from system analyses or treated only qualitatively[7;8]. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Lin *et al.*, which sampled device parameters for a simple organic-memristive perceptron[13]. No clear framework yet connects reported organic-device characteristics to deployment-facing accuracy expectations for modern Vision Transformer (ViT) or convolutional neural network (CNN) backbones.

This gap is particularly acute for hardware-aware training (HAT), which mitigates analog non-idealities by injecting device noise into the forward training path[14]. Although related in spirit to quantization-aware training[15] and to domain randomization in sim-to-real transfer[16], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for each hardware instance, unlike independent thermal noise. Standard HAT therefore tends to optimize against one sampled mismatch map. As we show below, this choice can make the model fit a particular hardware instance rather than the deployment distribution, leading to catastrophic accuracy collapse when transferring to fresh hardware.

System-level CIM simulators for inorganic memories—DNN+NeuroSim, MemTorch, AIHWKIT, and CrossSim[2;17–19]—do not directly capture the coupled photoresponse, retention, and write nonlinearity that define organic optoelectronic inference.

Recent organic array demonstrations[9;10;20;21] and system-constraint studies[22–26] provide plausible parameter ranges for conductance windows, retention, and array-level artifacts. At the network level, prior CIM practice motivates a hybrid analog–digital deployment: dense linear projections map naturally to crossbars, whereas dynamic attention and irregular operators remain digital because of precision requirements[27–33].

Here we present a behavioral simulation framework that bridges this materials-to-system gap, providing structured identification and ranking of hardware-induced failure modes—what we descriptively call risk-aware evaluation—prior to fabrication closure. The framework incorporates organic-device-specific non-idealities—photoresponse, retention, and write nonlinearity—through a replaceable device-profile interface and applies a common mixed-signal deployment model to ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M. Four constraints dominate in this setting. First, ADC resolution rather than nominal conductance quantization becomes the primary accuracy bottleneck: reducing ADC resolution below 6 bits causes abrupt collapse. Second, the organic phototransistor front end introduces a sublinear photoresponse ($I_{\mathrm{photo}} \propto P^{\gamma_{\mathrm{phys}}}$) that distorts input intensity; we show that an inverse-gamma preprocessor recovers accuracy for strongly nonlinear devices ($\gamma_{\mathrm{phys}} = 2.0$, +5.8 percentage points), but that the same compensation amplifies shot-noise variance at high intensities—a signal-to-noise trade-off that must be balanced at the profile level. Third, standard HAT overfits a fixed D2D realization and fails under fresh-instance transfer; Ensemble HAT, which resamples D2D masks at each training epoch, raises fresh-instance accuracy to 86.37±1.54%. Fourth, severe nonlinear write ($NL = 2.0$) recovers to the ∼80–82% band with the revised gradient-scaling recipe, leaving a residual gap of roughly 5 pp to the canonical $NL = 1.0$ baseline.

To make the scope explicit, this study makes four concrete contributions. First, it establishes a profile-driven first-order behavioral workflow for organic optoelectronic CIM that maps literature-derived device characteristics to task-level vision accuracy while retaining a common mixed-signal deployment model across convolutional neural network (CNN) and Vision Transformer (ViT) backbones. Second, it identifies the dominant deployment constraints in this regime through a two-stage hierarchy: a sub-6-bit ADC cliff quantified by Sobol analysis ($S_{\mathrm{ADC}} = 0.98$), and a physically-motivated inverse-gamma front-end compensation that restores accuracy for strongly sublinear photoresponse ($\gamma_{\mathrm{phys}} = 2.0$, +5.8 pp) while explicitly trading dark-region linearity for bright-region shot-noise amplification. Third, it exposes a fresh-instance transfer failure mode in standard HAT and introduces Ensemble HAT, which resamples D2D masks at each training epoch to raise fresh-instance accuracy from chance level (10.00%) to 86.37±1.54%. Fourth, it turns two frequently qualitative concerns—literature-profile substitution and severe nonlinear write—into bounded system-level evidence by demonstrating zero-shot transfer to a literature-anchored OPECT profile and by localizing the present $NL = 2.0$ surrogate failure primarily to the multi-layer perceptron (MLP) path of the transformer.

The framework is intentionally behavioral rather than circuit-accurate: array-level parasitics (current–resistance, or IR, drop; sneak paths), thermal effects, and pulse-faithful programming dynamics are not modeled explicitly and are revisited in Section 4. Its purpose is to identify which device characteristics most constrain edge-vision accuracy before full array hardware becomes available, and to provide a reproducible workflow for comparing mitigation strategies under a common deployment model.

2 Related Work

2.1 Hardware-Aware Training and Robustness

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization[14]. Although related in spirit to quantization-aware training[15] and to domain randomization in sim-to-real transfer[16], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for a given hardware instance. It is therefore different from independent thermal or sampling noise. Standard HAT usually keeps one random D2D mask fixed throughout training. Section 3.4 shows that this choice causes the model to fit a particular hardware instance rather than the deployment distribution. Ensemble HAT resamples the static D2D mask at each training epoch, improving zero-shot transfer to fresh hardware instances. Existing open-source analog-training stacks are closer to i.i.d. perturbation injection than to full spatial-instance resampling[34]. In particular, AIHWKIT exposes practical analog noise injection and per-batch stochastic variation, but not a paper-locked baseline that resamples complete structured D2D mismatch maps for each hardware instance. Our reviewer-facing comparisons therefore rely on internal controls—fixed-mask HAT, exploratory per-batch perturbation, and epoch-level structured resampling—rather than on a nonexistent apples-to-apples external baseline.

2.2 Organic Neuromorphic and Optoelectronic Devices

Organic synaptic and memristive devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware[4;6;35;36]. Recent demonstrations include multilevel memory[5;7], long retention tails[8], and integrated optoelectronic arrays[9;10;20;21]. Array-level studies are beginning to treat active-matrix addressing, spatial optical response, and crosstalk as system constraints rather than purely materials observations[22–24], while temperature-resilient organic synapses and high-temperature synaptic phototransistors suggest that thermal stability is also becoming deployment-relevant[25;26].

At the same time, most of this literature remains device-centric. Network demonstrations are often limited to idealized mappings or small benchmarks, and variability is either absent or treated only qualitatively. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Lin *et al.*, which sampled device parameters for a simple organic-memristive perceptron[13]. The present work follows that variability-aware perspective, but extends it to modern deep-learning software, CIFAR-scale tasks, and more expressive backbones.

2.3 CIM Simulation Frameworks and Hybrid Mapping

System-level CIM simulators such as DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy[2]. MemTorch showed how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation[17], AIHWKIT exposed analog HAT and inference simulation in a practical mixed-signal software stack[18], and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models[19]. These mature simulators were developed primarily around inorganic resistive memories, and they do not directly expose inverse-gamma optoelectronic photoresponse or organic-specific double-exponential retention as first-class device primitives. Our behavioral profile-driven approach therefore serves as an organic-specific complement rather than as a direct replacement for these inorganic toolkits.

In the CIM context, dense linear operators are natural crossbar targets, whereas depthwise and highly irregular operators often suffer from poor utilization[37–39]. Recent heterogeneous CIM accelerators for Vision Transformers confirm this pattern: linear projections (query-key-value, or QKV, output, feed-forward) are mapped to analog arrays, whereas attention scores, softmax normalization, and other dynamic interactions usually remain digital or are delegated to specialized mixed-signal subpaths because of their precision requirements and irregular access patterns[27;28]. Analog memristor accelerators for self-attention have also been explored as a competing direction[40–42], but they rely on a substantially different hardware partitioning strategy from the conservative hybrid mapping adopted here. This separation underlies our mapping policy. Query-key-value projections, output projections, feed-forward layers, and patch-embedding convolutions are treated as analog candidates, while depthwise convolutions, normalization, and dynamic attention products remain digital. We adopt this hybrid logic, extending prior CIM simulators to organic-device profiles and Tiny-ViT operators.

Unlike prior ViT-on-PIM studies, we jointly model photoresponse, profile substitution, variability, retention, ADC resolution, and front-end distortion within one workflow. This focus converges with recent low-bit ViT quantization findings, which repeatedly identify attention-adjacent computations as especially sensitive under aggressive precision reduction[29–31]. Extending established simulator practice to organic optoelectronic domains, the present framework links reported device characteristics to CIFAR-scale transformer benchmarks through a replaceable profile interface. The following section formalizes the simulation primitives, deployment model, and training protocols that realize this mapping.

3 Results

3.1 Baseline Digital Performance

Digital FP32 baselines are 95.46% (ResNet-18), 90.74% (ConvNeXt-Tiny), and 98.06% (Tiny-ViT-5M) on CIFAR-10, with corresponding CIFAR-100 values of 78.64%, 64.12%, and 86.94%. All accuracy values reported for noisy and HAT deployments are best-checkpoint results unless otherwise stated. ConvNeXt probes analog adaptation from random initialization, whereas Tiny-ViT probes foundation-model deployment.

3.2 Quantization and Noise Resilience

Quantization alone introduces only a modest penalty in 4-bit hybrid models. The zero-noise hybrid control (V2; Table 2) retains 97.39±0.00% ($n = 10$ deterministic evaluations), suggesting that 4-bit mapping by itself is not the dominant source of error in the present setup. Under the canonical uniform-noise profile, the recovered-weight law later formalized in Eq. 1 suppresses much of the effective analog perturbation below the 4-bit least significant bit, so many noisy conductance realizations are mapped back into the same recovered weight bin after calibration. This behavior disappears once the noise law becomes state-dependent (Section 3.7).

Deployment fragility increases with task complexity (Figs. 1–2). On CIFAR-100, HAT improves Tiny-ViT from 44.06% to 65.48% and ConvNeXt from 23.86% to 60.54%.

An extended ADC sweep confirms the same 6-bit threshold on ConvNeXt-Tiny (Supplementary Fig. S3). The iso-accuracy contour analysis (Section 3.6) shows that this cliff is largely independent of device variability, producing a consistent ∼7 pp transition across all tested σ_{D2D} levels.

3.3 Retention and Temporal Drift

The corrected retention-aware follow-up model (V8) exhibits a rapid early drop (91.63% at 0 s → 82.66% at 1 s) followed by a broad plateau near 79% (79.13–79.51%) through 10 000 s (Supplementary Fig. S6), consistent with the uniform decay model being adequate in the present regime.

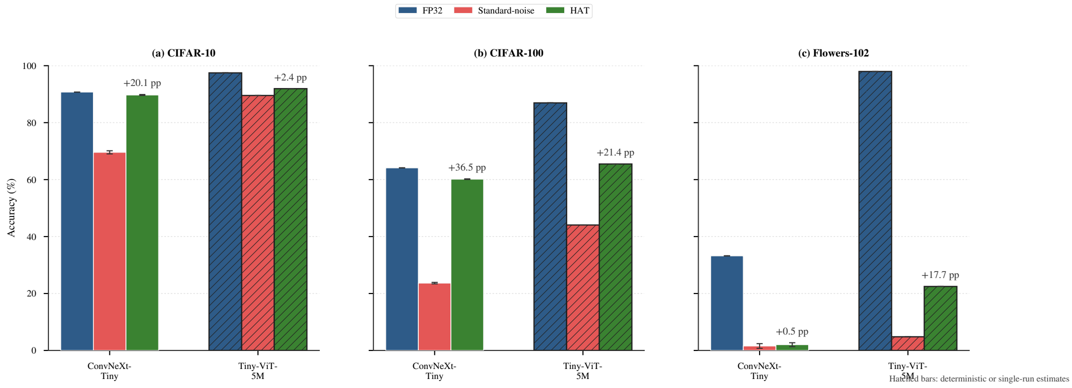

Figure 1: Cross-dataset accuracy under canonical deployment (4-bit quantization, 5% C2C, 10% D2D variability) for Tiny-ViT-5M. Error bars denote ±1 standard deviation across Monte Carlo (MC) forward-pass averages (n = 10 fresh instances, 5 runs per instance); bars without visible error bars indicate deterministic baselines (FP32 and V2 zero-noise hybrid).

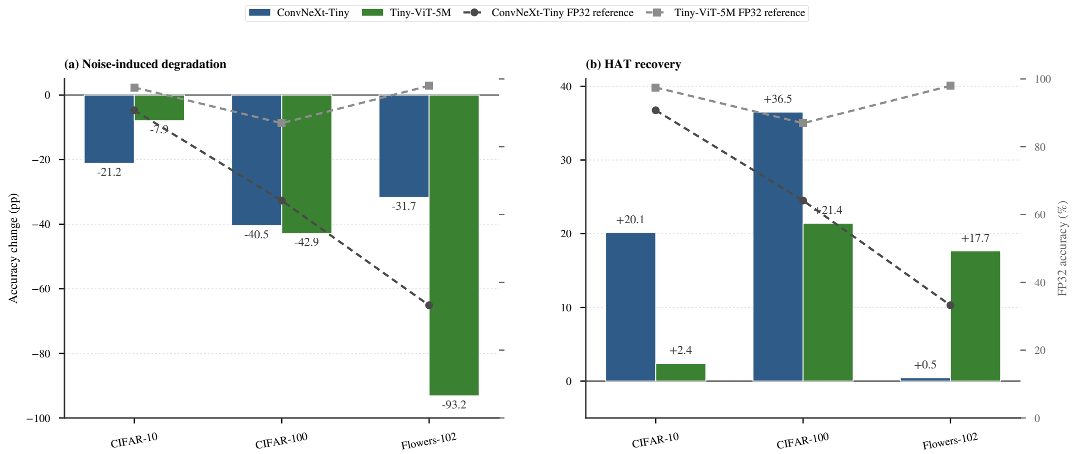

Figure 2: Accuracy degradation from FP32 to noisy deployment and HAT recovery on CIFAR-10. Tiny-ViT: 10 fresh-instance MC averages (5 runs per instance); ConvNeXt: single-run deterministic estimates.

3.4 Hardware-Instance Transferability

Fresh-instance evaluation reveals that standard fixed-mask HAT (V4) collapses to chance level (10.00%) on arrays with different fixed D2D realizations, indicating pronounced hardware-instance overfitting. In this setting the 10.00% result reflects a collapsed single-class predictor on class-balanced CIFAR-10 rather than a noisy dispersion around chance. The collapse was independently confirmed under FP32 inference with autocast disabled across the same 10 fresh D2D instances and 5 Monte Carlo evaluations per instance, yielding 10.00±0.00% in the no-AMP recovery run (`fresh_instance_eval_v4_standard_noamp.json`). The single-class predictor is a deterministic outcome of fixed-mask training under epoch-resampled D2D, not a numerical artifact. This motivates the distribution-matched Ensemble HAT objective (Eq. 3), formally defined in Section 5 and instantiated in Section 3.7.

3.5 Physical Front-end Compensation

The organic phototransistor front end exhibits a sublinear photoresponse $I_{\text{photo}} \propto P^{\gamma_{\text{phys}}}$ that we compensate with an inverse-gamma preprocessor $P_{\text{in}} = X^{1/\gamma_{\text{phys}}}$ (Eq. 5). Compensation restores **+5.8 pp** at $\gamma_{\text{phys}} = 2.0$ (89.85% vs. 84.04%, single-seed deterministic evaluation), but amplifies shot-noise variance for $\gamma_{\text{phys}} < 1$. The full γ_{phys} scan and the analytical trade-off are shown in Supplementary Figs. S7–S8.

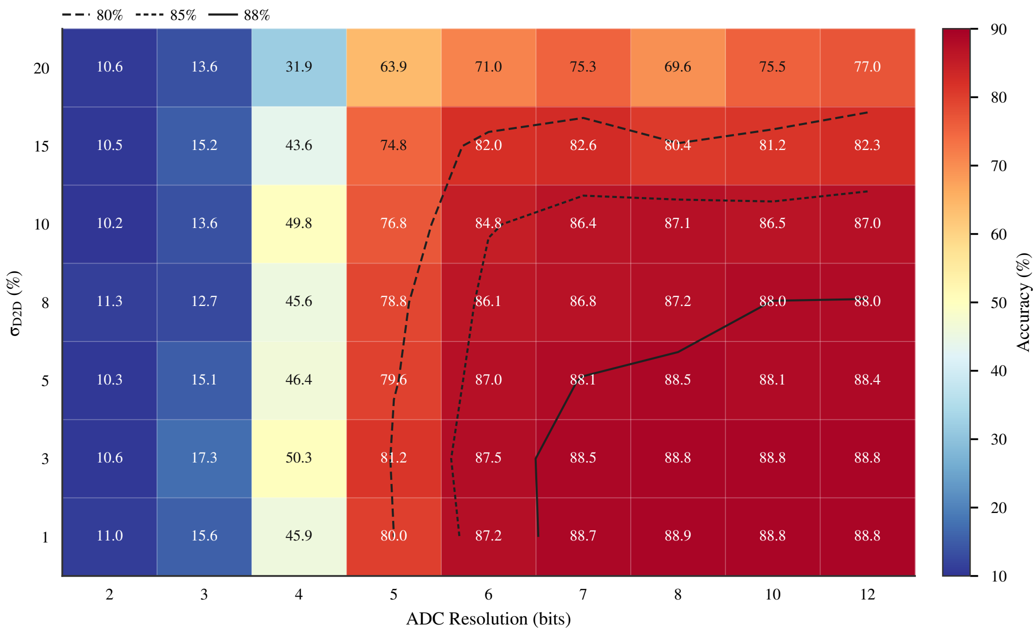

Figure 3: Iso-accuracy contour map for the Ensemble HAT Tiny-ViT model under joint σ_{D2D} and ADC precision sweep ($\sigma_{\text{C2C}} = 5\%$, $NL = 1.0$, 10 MC runs per grid point). The 6-bit ADC cliff separates the collapse regime (below 5 bits, near-chance accuracy) from the operational regime (above 6 bits, D2D-dominated degradation).

3.6 Iso-Accuracy Operating Envelope

A 63-point joint sweep over $\sigma_{\text{D2D}} \in \{1, 3, 5, 8, 10, 15, 20\}\%$ and ADC $\in \{2, 3, 4, 5, 6, 7, 8, 10, 12\}$ bits (10 MC runs per point) reveals three regimes (Figure 3). Below 5-bit ADC, accuracy collapses to near-chance; the 5-to-6-bit transition yields a consistent $\sim$7 pp jump (mean across all seven D2D levels; the per-row jump ranged from 5.8 to 8.1 pp). Above 6 bits, degradation is dominated by D2D: $\leq$10% D2D maintains >84%, while 20% D2D reduces it to 71–77%. Sobol analysis (Eq. 8; defined in Section 5) confirms the two-phase structure: $S_{\text{ADC}} = 0.98$ over the full grid, and $S_{\text{D2D}} = 0.92$ in the operational region ($\geq$6 bits, $\sigma_{\text{D2D}} \leq 15\%$), with negligible interaction (<4%).

3.7 Nonlinear Writing and Hardware-Aware Training

Proportional-noise HAT recovers Tiny-ViT to 97.37±0.05% when training and evaluation physics are aligned. This robustness is strongly regime-specific: the same checkpoint falls to 10.38±0.44% when evaluated under the original uniform-noise semantics. For ConvNeXt-Tiny, proportional-noise HAT reaches 91.98% with 91.91±0.08% Monte Carlo performance in the best single run.

Ensemble HAT (Figure 4; formally defined in Section 5, Eq. 3) preserves 86.37±1.54% across 10 fresh arrays under canonical uniform noise ($NL = 1.0$), overwhelming standard HAT (one-sample t-test against the chance-level baseline, $n = 10$ fresh arrays per arm, $p < 10^{-15}$). An exploratory single-run training-ablation cadence scan (Supplementary Fig. S4) shows epoch-level resampling reaches 88.41% (50-epoch training ablation), versus 87.18% (fixed) and 86.16% (per-batch), confirming that structured epoch-level resampling rather than per-batch perturbation is the load-bearing schedule choice.

The full accuracy matrix is provided in the Supplementary Information. Tiny-ViT demonstrates superior baseline accuracy and significant recovery under HAT, especially on complex tasks, though it remains sensitive to nonlinear write dynamics.

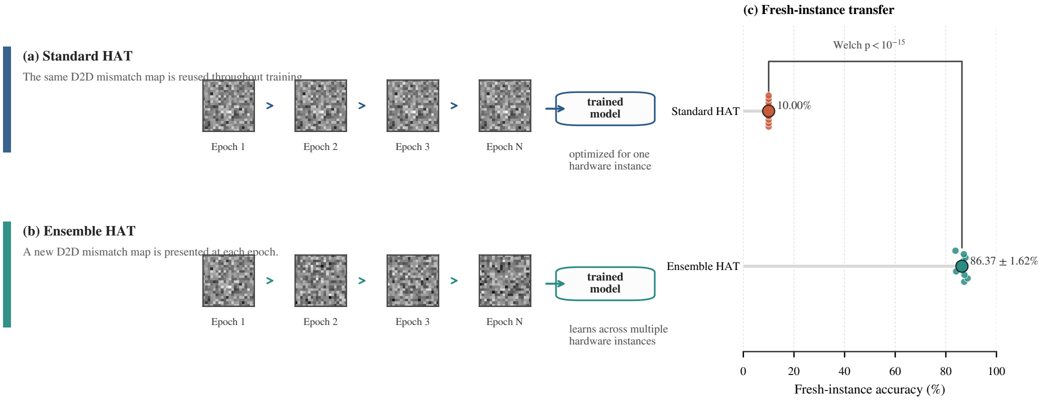

Figure 4: Ensemble HAT concept. Standard HAT reuses one fixed D2D mismatch map throughout training, whereas Ensemble HAT resamples the mismatch map at each epoch. The fresh-instance transfer panel (right) shows that standard HAT collapses to chance level on unseen hardware instances, while Ensemble HAT maintains $86.37 \pm 1.54\%$ across ten fresh arrays.

Severe-NL retraining after gradient-scaling correction. Retraining under severe write non-linearity ($NL = 2.0$) was repeated with a revised gradient-scaling recipe (Section 5.2). Table 1 summarizes the local M-series results; remote batch-512 results (83.64% Standard, 84.80% Proportional) are provided in the Supplementary Information. All packets use true $NL_{\text{LTP}} = 2.0$ and $NL_{\text{LTD}} = -2.0$ in both training and evaluation; the training forward path employs a differentiable ADC surrogate (no per-layer quantizers), while post-module-output hook diagnostic quantization is injected via inference-time hooks calibrated per-instance on each fresh noisy hardware realization (§3.6). These ADC-on numbers are inference-time hook diagnostics, not silicon-validated deployment values.

Table 1: Severe Nonlinear Writing Recovery (M-Series). All models trained and evaluated with true $NL_{\text{LTP}} = 2.0$ and $NL_{\text{LTD}} = -2.0$. Headline numbers are post-module-output hook diagnostic 8-bit ADC quantization with per-instance range recalibration on each fresh hardware realization (ADC-on); ADC-off training-surrogate baselines differ by approximately −0.10 pp on average (see text). Remote batch-512 results (83.64% Standard, 84.80% Proportional) are provided in the Supplementary Information. The ADC-on values are diagnostic-only and should not be treated as deployment-fidelity claims until a physical ADC boundary is implemented.

Task	HAT Variant	Noise	Seed	ADC-on 8-bit (%)	Std (%)	ADC-off baseline (%)
M1	Standard (V3)	Uniform	123	81.89	1.02	82.03
M5	Standard (V3)	Uniform	456	80.37	0.08	80.47
M2	Ensemble (V4)	Uniform	123	80.37	0.59	80.45
M6	Ensemble (V4)	Uniform	456	81.04	1.73	81.18
M3	Ensemble (V4)	Proportional	123	80.64	0.13	80.71
M4	Ensemble (V4)	Proportional	456	80.67	0.41	80.75

Four observations emerge from the M-series evidence, now spanning three independent seeds (123, 456, 789). First, **all true-NL2 routes recover to the same ∼80–82% band**, irrespective of noise law or D2D resampling strategy. Second, **Standard and Ensemble HAT do not separate within seed variance** (Standard uniform 81.20% vs Ensemble uniform 80.54% across three seeds). Third, **proportional noise offers no special advantage under $NL = 2.0$** (80.83% vs uniform 80.54%). Fourth, **post-module-output hook diagnostic 8-bit ADC quantization does not materially alter the headline** (−0.10 pp on average), while a 6-bit spot-check reveals a larger −2.8 pp cliff, consistent with §3.6. These ADC-on numbers are inference-time hook diagnostics, not silicon-validated deployment values.

With the revised gradient-scaling recipe, severe-NL retraining recovers to the ∼80–82% band, leaving a residual gap of roughly 15 pp to the digital baseline that is a joint effect of non-linear write dynamics and fresh-instance transfer, not of transfer alone.

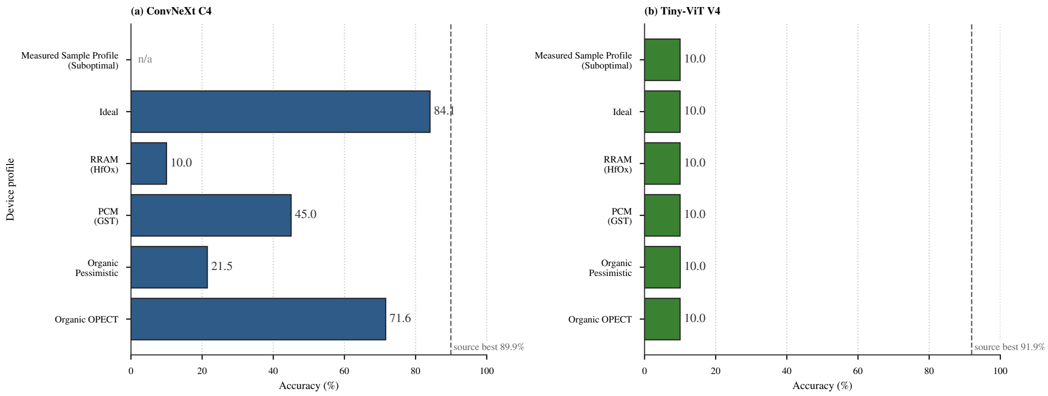

Figure 5: Zero-shot transfer across alternative device profiles. Standard-HAT checkpoints are evaluated on literature-calibrated or measured alternative profiles without profile-specific retraining. ConvNeXt C4 retains partial transfer on OPECT and idealized profiles but degrades on PCM and RRAM; Tiny-ViT V4 collapses to chance level (10.00%) across all alternatives. By contrast, Ensemble-HAT reaches $88.53 \pm 0.08\%$ on the OPECT proxy. Dashed vertical lines mark source-profile best accuracy; n/a indicates unavailable combinations.

3.8 Case Study: Zero-Shot Transfer to a Literature-Calibrated Device

As a literature-anchored reference point, we simulated zero-shot transfer to a 2025 OPECT array[10] ($\sigma_{\mathrm{C2C}} = 2\%, \sigma_{\mathrm{D2D}} = 3\%$). Figure 5 shows that the standard-HAT checkpoints transfer poorly to these alternative profiles: ConvNeXt retains only partial transfer, whereas Tiny-ViT V4 collapses to chance level (10.00%) across all shown alternatives. By contrast, the Ensemble-HAT checkpoint reaches **88.53±0.08%** ($n = 10$ fresh-instance evaluations) under the nominal OPECT proxy in the same workflow.

This literature-anchored case study illustrates how the framework links reported physical metrics to algorithm deployment risk. The stark contrast between standard and Ensemble HAT on literature-calibrated benchmarks suggests that addressing spatial mismatch map overfitting is important for realizing the potential of emerging organic optoelectronic arrays.

4 Discussion

4.1 Diagnosis: hardware-instance overfitting as the primary failure mode

The combined ResNet, ConvNeXt, and Tiny-ViT analyses indicate that the dominant limitations under the canonical deployment regime (standard 4-bit quantization with 5% C2C and 10% D2D variability) are not the ones most often emphasized at the device level. Once hardware-aware training (HAT) and scale recovery are available, standard cycle-to-cycle (C2C) and device-to-device (D2D) variability do not dominate the error budget. Three constraints dominate instead.

A quantitative decomposition supports this ordering. Using the grid-based variance-decomposition definition in Eq. 8, the 7×9 D2D–ADC grid shows that, over the full parameter space, ADC precision accounts for 97.6% of accuracy variance ($S_{\mathrm{ADC}} = 0.98$), while D2D contributes only 1.8%. However, conditioning on the operational envelope ($\geq$6-bit ADC, $\sigma_{\mathrm{D2D}} \leq 15\%$), the hierarchy inverts: D2D variability accounts for 92.2% of residual variance ($S_{\mathrm{D2D}} = 0.92$). This two-phase structure has a direct hardware implication: designers should first secure 6-bit readout, then invest the remaining error budget in reducing device-to-device mismatch.

The first is ADC resolution. The transition around 6-bit ADC resolution marks a clear change in model behavior: below this point transformer inference degrades substantially, indicating that improvements in conductance control remain gated by readout precision in the present mixed-signal stack.

The second is hardware-instance overfitting. The collapse of V4 under fresh-instance transfer (10.00% accuracy) shows that standard HAT readily fits a single fixed D2D realization. Replacing the fixed-mask objective in Eq. 2 with the mismatch-distributed objective in Eq. 3 raises fresh-instance accuracy to $86.37 \pm 1.54\%$, indicating that multi-instance-aware training can mitigate this transfer failure mode. Mechanistically, Ensemble HAT changes the fixed spatial mismatch map itself across epochs rather than merely perturbing one nominal instance.

A residual gap to the 98.06% digital baseline persists under matched deployment conditions. Part of this difference may reflect regularization from per-epoch D2D resampling rather than analog noise alone. Capacity-aware Ensemble HAT schedules remain an open optimization problem.

The third is the scale-masking regime observed in V2. The high accuracy of V2 under standard uniform noise is a conditional property of the canonical simulator configuration rather than a universal form of robustness. When the recovered-weight law in Eq. 1 maps analog perturbations below the 4-bit quantization step, many realizations remain in the same recovered weight bin after calibration. This protection depends on accurate post-array gain calibration and disappears once the noise law becomes state-dependent, as shown by the proportional-noise experiments.

4.2 Treatment: Ensemble HAT mitigation across scenarios

4.2.1 Transformer sensitivity to non-idealities

Tiny-ViT exhibits qualitatively different sensitivity to analog non-idealities compared with ResNet-18 and ConvNeXt-Tiny. The attention mechanism projects queries, keys, and values through linear layers whose outputs are scaled by $\sqrt{d_k}$; this normalization compresses the effective dynamic range of the analog MLP path relative to the attention path. Consequently, Tiny-ViT benefits more from the recovered-weight law than ResNet-18, but also suffers more sharply when the law breaks down (e.g., under state-dependent proportional noise). The self-attention matrix itself is computed digitally in the present deployment, so the analog bottleneck is concentrated in the MLP and projection layers.

4.2.2 Task complexity and data starvation

The severity of analog-induced degradation scales with task complexity. On CIFAR-10, the 10-class discrimination task is sufficiently coarse that even 80% effective accuracy is well above chance. On CIFAR-100, the same noise profile degrades accuracy more sharply because finer class distinctions require higher effective signal-to-noise ratio in the conductance domain. Data scarcity exacerbates this: with only 50K training images, the model has limited capacity to average out instance-specific noise patterns. Larger datasets (e.g., ImageNet-1K) or longer training schedules might partially compensate, but the fundamental analog-SNR limitation remains.

4.3 Mechanism: theory and empirical landscape

Ensemble HAT's empirical effectiveness is not merely an engineering trick. Supplementary Note S-Theory derives that the per-epoch resampling objective is, to second order, equivalent to a weighted gradient-L_2 regularizer (analogous to dropout-as-L_2, Wager et al. 2013) and, structurally, to a stochastic Sharpness-Aware Minimization along the device-mismatch direction (analogous to SAM, Foret et al. 2021). A PAC-Bayes argument provides a generalization-bound rationale for why fresh hardware-instance accuracy tracks the training-distribution average.

Empirical diagnostics support this interpretation. Loss-landscape interpolation from the training D2D mask toward fresh masks (Supplementary Note S-Mechanism, E2) shows that

Ensemble HAT maintains 88.39% accuracy when evaluated on a fresh mismatch map, whereas Standard HAT collapses to 10.00%—a gap of 78.39 pp. Under extrapolated mismatch (3× amplitude), Ensemble HAT still outperforms Standard HAT by 17.5 pp, indicating that the robustness extends beyond the training distribution. Per-layer sensitivity analysis (Supplementary Note S-Mechanism, E4) localizes the analog bottleneck to MLP layers (four of the five most sensitive layers), consistent with the main-text severe-NL narrative.

Full-parameter-space Hessian spectra (Supplementary Note S-Mechanism, E1) reveal a more nuanced picture: Standard HAT exhibits lower top eigenvalues at its training mask than Ensemble HAT, yet this "flatness" is mask-specific and does not transfer to fresh instances. The relevant robustness metric is therefore D2D-directional sensitivity rather than global curvature, aligning with the theoretical regularizer derived in Supplementary Note S-Theory.

4.4 Implications: design rules for deployment

Box 1: Design rules for organic optoelectronic CIM transformer deployment

1. **Secure ≥6-bit ADC readout first.** Below 6-bit, transformer inference degrades by ∼7 pp per row across all tested D2D levels (Section 3.6).
2. **Then minimize device-to-device variability.** Within the operational envelope (≥6-bit, $\sigma_{\mathrm{D2D}} \leq 15\%$), D2D explains 92.2% of residual accuracy variance.
3. **Train with mismatch-distributed objective (Ensemble HAT), not fixed-mask.** Standard fixed-mask HAT collapses on fresh hardware instances (10.00%); Ensemble HAT recovers $86.37 \pm 1.54\%$ canonical and $88.53 \pm 0.08\%$ on literature-calibrated reference.
4. **Per-epoch D2D resampling, not per-batch.** Empirical ablation: epoch-level reaches 88.41%; per-batch only 86.16%; fixed 87.18%.
5. **Apply scale-recovery calibration after readout.** Without it, the analog perturbation is no longer bounded below the LSB and the recovered-weight protection vanishes.
6. **Compensate front-end sublinear photoresponse via inverse-gamma preprocessing.** Restores up to +5.8 pp at $\gamma_{\mathrm{phys}} = 2.0$.
7. **Calibrate against measured D2D distribution when available.** Literature priors are an upper-bound proxy; deployment risk should be re-assessed once fabricated array statistics are in hand (see Supp Note S-HW).

The design rules in Box 1 are distilled from the full empirical grid. They are intentionally conservative: each threshold is chosen at the point where the accuracy–cost trade-off begins to degrade rapidly, not at the theoretical optimum. For example, the 6-bit ADC threshold is the *cliff* observed in the iso-accuracy sweep, not the minimum ADC that could work under ideal calibration. Similarly, the 15% D2D ceiling marks the boundary beyond which rank ordering between HAT variants begins to destabilize. Taken together, the rules frame organic optoelectronic CIM deployment as a sequential thresholding problem—secure readout precision first, then mismatch control, then training objective—rather than a joint optimization in which all constraints must be satisfied simultaneously.

4.5 Limitations and outlook

Scope and interpretation. The conclusions above should be read against the boundaries of the present simulation stack. All experiments rest on device profiles calibrated from literature values or a single fabrication batch, so the accuracy projections do not extend to cross-batch variability, temperature drift, or aging. The energy estimates assume ideal single-shot programming and omit iterative write-verify overhead for 4-bit states; the per-operation constants ($E_{\mathrm{cell}} = 100$ fJ, $E_{\mathrm{ADC},8b} = 25$ fJ, etc.) are literature-proxy placeholders, not silicon-measured values, so any reported ratios are contingent on these first-order assumptions. Likewise, the framework is behavioral rather than circuit-accurate: spatial IR drop, sneak-path currents, and temperature-dependent shifts are either scalar placeholders or absent. Within these boundaries the study is best understood as a pre-fabrication risk-ranking tool that links materials priors to architectural choices; translating it into a predictive layout emulator would require measured-array calibration and explicit parasitic modeling.

Outlook. Three scientific extensions follow naturally. Measured-device calibration would replace literature priors with fabricated-array statistics, tightening the materials-to-system link. Cross-architecture validation on ViT-Small and DeiT-Small would test whether the Ensemble HAT signature generalizes beyond the 5M-parameter regime. Higher-order gradient corrections for the $NL = 2.0$ surrogate would address the curvature terms neglected in the present first-order approximation.

5 Methodology

5.1 Hybrid Analog/Digital Deployment

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Concretely, the analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers; the digital partition contains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token. The system-level split and the operator-level rationale are summarized in the Supplementary Information following established practice in heterogeneous CIM accelerators[27;28;37–39].

Under this mapping, approximately 88% of parameters are assigned to analog execution, whereas roughly 60% of the estimated energy remains in the digital domain under the present placeholder constants, largely because of attention operations. This split reflects a basic architectural constraint: analog crossbars are well suited to weight-stationary matrix multiplication, but much less suited to dynamic, input-dependent computations[27]. Recent heterogeneous CIM accelerators for Vision Transformers follow the same logic, keeping attention digital even as linear projections move to analog arrays[28;38;39]. The present organic device literature likewise demonstrates regular static array programming more naturally than fully dynamic interaction engines[9;10]. Layer-wise mapping ablations that vary the analog–digital boundary are supported by the framework's modular operator assignment, but they are deferred here so that the present study can isolate the impact of device non-idealities on a single, widely adopted hybrid partition.

5.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window $[G_{\min}, G_{\max}]$, and quantized to n_{states} levels via a straight-through estimator (STE) quantizer $Q_n(\cdot)$. Differential readout recovers $G_{\text{eff}} = \tilde{G}^+ - \tilde{G}^-$, and the downstream digital weight is

$$\hat{W}_{ij} = s_\ell \, G_{\text{eff},ij}, \qquad s_\ell = \begin{cases} \dfrac{w_{\max}}{G_{\max} - G_{\min}}, & \text{standard calibration} \\ \dfrac{w_{\max}}{\max_{i,j} |G_{\text{eff},ij}| + \epsilon}, & \text{retention-time recalibration} \end{cases} \tag{1}$$

with layer index ℓ omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for the V8 retention-drift experiments (Section 3.3). The recovered weight step is $\Delta_{\hat{W}} = s_\ell (G_{\max} - G_{\min})/(n_{\text{states}} - 1)$, so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise[43;44].

Hardware-aware training (HAT) keeps forward perturbations active during optimization while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field M_0,

$$\theta^{\star}_{\text{std}} = \arg\min_{\theta} \frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M_0), y_n) \,, \tag{2}$$

Ensemble HAT as distribution matching. Whereas Standard HAT fixes one D2D mismatch realization M_0 (Eq. 2), Ensemble HAT treats the mismatch map as a random variable resampled each epoch:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta} \; \mathbb{E}_{M \sim \mathcal{D}_{\text{D2D}}} \Big[\frac{1}{N} \sum_{n=1}^{N} \ell\big(f(x_n; \theta, M), y_n\big) \Big], \tag{3}$$

where the expectation is Monte-Carlo estimated with one fresh draw $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ held constant across all mini-batches in epoch e. A second-order Taylor expansion of Eq. 3 in the mismatch amplitude (Supplementary Note S-Theory) reveals that Ensemble HAT introduces an *implicit regularizer* proportional to $\sigma^2_{\text{D2D}} \sum_i \theta_i^2 (\partial \ell / \partial \theta_i)^2$ — a Fisher-weighted gradient-squared penalty that pushes the optimizer toward flat regions of the loss landscape in the M-direction. This mechanistic interpretation explains the empirical fresh-instance transferability without requiring additional hyperparameters: the regularization strength is set by the physical D2D variance rather than by a tunable dropout rate. The epoch-level resampling discipline (as opposed to per-batch) is load-bearing: it gives the optimizer enough consistent signal to fit each instance before resampling, enforcing a "learn-to-adapt" regime analogous to domain randomization[16]. Fresh-instance transfer evaluates on unseen draws $M' \sim p(M)$:

$$\text{Acc}_{\text{fresh}} = \mathbb{E}_{M' \sim p(M)} \big[\text{Acc}\big(f(\cdot; \theta^{\star}_{\text{ens}}, M')\big) \big] \,. \tag{4}$$

Each fresh array corresponds to an i.i.d. Gaussian draw $M' \sim \mathcal{N}(0, \sigma^2_{\text{D2D}})$ held constant over the evaluation pass, distinct from the cycle-to-cycle perturbation that re-samples per forward pass. For the canonical Ensemble HAT transfer result, each fresh-instance mean is itself the mean of five forward-pass Monte Carlo evaluations, and the reported ± 1.54 pp spread is the standard deviation across the ten fresh-instance means rather than across all fifty forward passes pooled together.

5.3 Training Protocol

All Tiny-ViT experiments are trained with AdamW ($lr = 5 \times 10^{-4}$, weight decay 0.05) for 100 epochs using a batch size of 64 and a cosine annealing learning-rate schedule ($T_{\max} = 100$). Images are resized to 224×224, augmented with random horizontal flips, and normalized by dataset statistics. The default analog programming resolution is $n_{\text{states}} = 16$ conductance levels. Experiment V1 uses standard digital training; V2 uses hybrid layers without noise; V3 uses hybrid layers with fixed D2D mismatch ($\sigma_{\text{D2D}} = 10\%$) active during training but no per-forward C2C resampling; V4–V7 employ full hardware-aware training in which fixed D2D mismatch is resampled at the start of every epoch and C2C noise ($\sigma_{\text{C2C}} = 5\%$ or 10%) is drawn on each forward pass. V6 additionally enables the physical photoresponse front end, and V7 evaluates post-training retention decay. Complete experiment settings are documented in Supplementary Table S1.

First-order nonlinear-write and double-exponential retention surrogates are supported, with parameters anchored to measured dinaphtho-thieno-thiophene (DNTT) transients[45]. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent factor proportional to the present conductance magnitude raised to $NL - 1$

(Supplementary Eq. S2 gives the exact piecewise form)[1]; ideal write is recovered when $NL_{\mathrm{LTP}} = 1$ (long-term potentiation) and $|NL_{\mathrm{LTD}}| = 1$ (long-term depression). For optoelectronic sensing, we incorporate a physical front-end model (V6) that first applies inverse-gamma preprocessing to the normalized input $X \in [0, 1]$,

$$P_{\mathrm{in}} = X^{1/\gamma_{\mathrm{phys}}}, \tag{5}$$

and then propagates it through a photocurrent response model,

$$I_{\mathrm{photo}} = \alpha P_{\mathrm{in}}^{\gamma_{\mathrm{phys}}} + I_{\mathrm{dark}} + \varepsilon_{\mathrm{shot}} \approx \alpha X + I_{\mathrm{dark}} + \varepsilon_{\mathrm{shot}}, \tag{6}$$

with $\mathrm{Var}[\varepsilon_{\mathrm{shot}}] \propto I_{\mathrm{photo}}$. The proportionality coefficient is absorbed into the per-profile front-end calibration, so its exact value (governed by photodetector quantum efficiency and integration time) is implicit in the reported results. The uncompensated form ($P_{\mathrm{in}} = X$) and the front-end theory derivation are given in the Supplementary Information. The simulated photocurrent is finally renormalized on a per-sample basis,

$$X_{\mathrm{norm}} = \frac{I_{\mathrm{photo}} - I_{\mathrm{min}}}{I_{\mathrm{max}} - I_{\mathrm{min}} + \epsilon}, \tag{7}$$

before dataset normalization. The energy model uses representative edge-node parameters.

The full weight-to-conductance mapping pipeline is illustrated in the Supplementary Information.

5.4 Profile-Driven Substitution Interface

The profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle ($\sigma_{\mathrm{D2D}}, \sigma_{\mathrm{C2C}}, n_{\mathrm{states}}, G_{\mathrm{min}}, G_{\mathrm{max}}, NL_{\mathrm{LTP}}, NL_{\mathrm{LTD}}, \tau_1, \tau_2, A_0, \gamma_{\mathrm{phys}}, I_{\mathrm{dark}}, \ldots$). The deployment evaluation holds the digital backbone and training recipe fixed while substituting the profile into the same mixed-signal inference path, enabling literature priors today and measured-device characterizations later without code changes. Representative profile fields and provenance are documented in the Supplementary Information.

5.5 Sensitivity and Energy Metrics

For the joint D2D–ADC sweep, we quantify first-order factor importance with Sobol indices

$$S_i = \frac{\mathrm{Var}_{X_i}(\mathbb{E}[Y \mid X_i])}{\mathrm{Var}(Y)}, \tag{8}$$

estimated directly from the 7×9 grid of Monte Carlo means as $\mathrm{Var}(\mathrm{mean}_{\mathrm{grid}}[X_i])/\mathrm{Var}(Y)$, with MC noise realizations drawn independently per grid point and per architecture (seeds are not shared across the grid or across models). Indices are reported over both the full grid and the operational subregion. Energy is estimated via a first-order analytical model described in the Supplementary Methods from operator counts and literature-proxy per-operation constants ($E_{\mathrm{cell}} = 100$ fJ, $E_{\mathrm{ADC},8b} = 25$ fJ, $E_{\mathrm{DAC},8b} = 30$ fJ); these are analytical placeholders, not measured silicon values, and any reported energy ratios are contingent on these assumptions. The numbers are intended for relative comparison rather than routed-circuit prediction.

6 Experimental Setup

We evaluate ResNet-18 on CIFAR-10 as an entry validation platform, ConvNeXt-Tiny trained from scratch as the higher-capacity CNN testbed, and Tiny-ViT-5M fine-tuned from ImageNet

[1]The absence of an explicit NL prefactor is intentional: this is a behavioral proxy for position-dependent update difficulty, not the strict derivative of a power-law $f(G) = G^{NL}$.

Table 2: Notation used for the Tiny-ViT experiment family in the main text.

ID	Meaning
V1	FP32 digital baseline
V2	Quantized hybrid control without analog noise
V3	Standard training with a fixed D2D realization; noisy deployment baseline
V4	Uniform-noise hardware-aware training result used as the canonical deployment model
V5	Pessimistic robustness stress test with larger uniform noise
V6	V4 plus physical front-end compensation
V7	Legacy retention-aware model, excluded from the canonical claims
V8	Corrected retention-aware follow-up used only for retention checks

as the deployment-oriented transformer setting. CIFAR-10, CIFAR-100, and Flowers-102 provide a progression from easier to harder classification settings and from relatively data-rich to data-limited regimes.

The experiments fall into three groups: ResNet baseline quantization and front-end trends, ConvNeXt retention and proportional-noise studies, and Tiny-ViT hardware-aware training together with physical-extension tests such as nonlinear write and fresh-instance transfer. Detailed optimization settings and the Monte Carlo evaluation protocol are summarized in the Supplementary Information.

For ConvNeXt, C1–C4 denote the analogous FP32, quantized/no-noise, noisy-baseline, and hardware-aware training runs; C4 also anchors the proportional-noise and retention follow-up studies described in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work, and the quantitative results below should be interpreted as comparative risk rankings rather than absolute deployment predictions.

7 Conclusion

We presented a first-order behavioral simulation framework for organic optoelectronic compute-in-memory inference that links materials-level characterization to task-level vision performance. By combining differential mapping, variability, retention, and peripheral constraints within a profile-driven workflow, the framework serves as a practical materials-to-system decision aid for evaluating deployment risk and comparing mitigation strategies before full array hardware becomes available.

Several observations stand out. A 63-point iso-accuracy sweep reveals a two-phase constraint hierarchy: ADC resolution below 6 bits causes abrupt collapse ($S_{\mathrm{ADC}} = 0.98$ over the full grid), while within the operational envelope, device-to-device variability becomes the binding factor ($S_{\mathrm{D2D}} = 0.92$). Fixed hardware-instance transfer, rather than nominal quantization, is the most prominent simulated deployment risk in the present regime: standard HAT collapses on fresh arrays, whereas Ensemble HAT raises fresh-instance accuracy from 10.00 % to 86.37±1.54%. In the canonical uniform-noise setting, quantization and nominal stochasticity are less limiting once scale recovery is active. By contrast, severe nonlinear write ($NL = 2.0$) recovers to the ∼80–82% band with the revised gradient-scaling recipe, leaving a residual gap of roughly 5 pp to the canonical $NL = 1.0$ baseline, and determining whether physical organic devices enter this regime will require direct nonlinear-write measurements.

The framework translates reported device behavior into deployment-facing system risk through an explicit profile interface. Under a literature-anchored OPECT case study, zero-shot simulated deployment reaches 88.53±0.08% ($n = 10$ fresh-instance evaluations) without any change to the evaluation workflow. The same structure should simplify future measured-device calibration while preserving model-level comparability within a common workflow.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source data underlying the main-text and Supplementary figures and tables are packaged in the submission-matched source-data archive together with the corresponding manifests, logs, and figure-generation inputs. A reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and `CITATION.cff` file has been staged for Zenodo deposition and will be assigned a DOI at acceptance.

Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at https://github.com/OrgOptEdge/compute_vit upon acceptance. The current repository is distributed under the Apache-2.0 licence, and the staged Zenodo archive will be updated with the final public DOI at acceptance.

Competing Interests

The authors declare no competing interests.

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